UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fifth Semester – Spring 2024

Roll No.

Paper: Computational Physics - I | Course Code: PHYS-3502 | Time: 3 Hrs. | Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions: (6x5=30)

i. Write C++ program to print the series and also display corresponding sum for the even numbers in the range from 2022 to 2300 for three iterations

ii. Explain with examples a) while loop b) main() and c) sqrt()

iii. Write C++ program to calculate and print area and circumference of a circle. Take radius values from the user. Implement with arrays.

iv. Write C++ program to calculate $Q = p^3 - 55p^2 - 37p + 46$ and print values for Q and p. Also find average of Q values. Take p values from the user for 10 iterations.

v. Write program to determine equivalent of n capacitors connected in series and parallel. Take capacitor values from the user.

vi Write program to calculate and print force value by taking inputs for mass and acceleration values from the user for 10 iterations.

Answer the following questions.

Q.2.	Write C++ program to evaluate the $\int_1^6 \frac{1}{x^3-5}dx$ by both Trapezoidal's Rule and by Simpson's rule (Use n=6). Write C++ program which reads in a value in meters and convert it into kilometer by user function for 5 iterations.	7+3
Q.3.	Suppose A and B be 3x3 matrices. Write C++ program which reads in entries of the matrices and calculate (i) **C=A + B**, (ii) **D=AxB^t** and (iii) **E=22A - 20B** (iv) sum of first rows of C and E and (v) sum of A and D. Write C++ program to read 30 values each into the linear arrays a and b. Print the array c= a + b. Print d = 0.2a * 0.3b.	7+3
Q.4.	Write C++ program for the Simple harmonic motion (SHM) of a mass spring system by Euler's method under the following conditions: (g=9.8 m/s^2, initial position zero and velocity 15 m/s, time step 0.1 sec. and maximum time 15 sec., k = 1 N/m, m=1kg, damping coefficient = 0.5 N/ms. ω =0.01 s^{-1} and f_0=1.5N.). Calculate and print values for time, position, velocity and acceleration. Also suggeste change to implement program for Forced H.M and Damped. H.M. Write C++ program to calculate and print the sum S such that: $S = \frac{1}{k}\sum_{i=1}^{k} a_i \bullet b_i$. Take k=10 and read values of a and b from the user.	7+3

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B.S. 4 Years Program / Fifth Semester – Fall 2023

Roll N

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iv. Write C++ program to calculate $Q = p^3 - 55p^2 - 37p + 46$ and print values for Q and p. Also find average of Q values. Take p values from the user for 10 iterations.

v. Write program to determine equivalent of n capacitors connected in series and parallel. Take capacitor values from the user.

vi Write program to calculate and print force value by taking inputs for mass and acceleration values from the user for 10 iterations.

Q.2. Answer the following questions. (3x10=30)

I. Write a C++ program to calculate and print time against position values for Damped Harmonic Oscillator by using Euler's Method for: $a = -kx - {}^{bv}/_{m}$, given that $g = 9.8\,m/s^2$, Initial position=x=0, Initial velocity=v_i=50m/s, Initial time=t=0 s, h=0.15 s, t_{maxi}=5.5s, mass=m=1kg, k=, b=0.5.

II. Write a C++ program to find roots of equation $2x - Cos\,x - 3$ at $x_0 = \frac{\pi}{2}$ using Simple Iterative method (one of numerical methods)

III. Write a C++ program to evaluate the integral I= $\int_0^1 e^x\,dx$ using Simpson's 1/3 rule for the range [0,1] and h=1/6.

UNIVERSITY OF THE PUNJAB

BS (After Associate Degree) : Fifth Semester – Fall 2023

Paper: Computational Physics – I Course Code: PHYS-3502

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SH[illegible]

Q.1. Solve the following questions. (6x5=30)

i. Write C++ program print v and s such that $v=u+at$ and $s= ut+\frac{1}{2} at^2$ take input from user for three iterations.

ii. Explain with examples a) for loop b) int and c) pow()

iii. Write program to calculate A and V such that $A= 4 \pi r^2$ and volume $V=4/3 \pi r^3$ Take radius values from the user. Implement with arrays.

iv. Store and print 9 random numbers into a matrix A (3x3). Also print A values.. Find and print transpose of A values.

v. Write program to print x vs y such that $y(x) = x^4+3x^3-7x-28$ Take x values from the user for 10 iterations.

vi. Write program to calculate and print force value by taking inputs for x and k values from the user for 10 iterations.

Solve the following questions. (3x10=30)

Q.2.	Write C++ program to evaluate the integral $\int_1^6 (\frac{\sqrt{x}+1}{3})dx$ by both trapezoidal rule and Weddle's rule (with n=6) Write C++ program to generate twenty one random-number pair of points (X,Y) and calculate 20 distance values of consecutive points. Discuss how these points show Brownian motion?	6+4
Q.3.	Suppose G and H be 3x3 matrices. Write C++ program which reads in entries of the matrices and calculate (i) J=2G + H, (ii) K=GxH (iii) L=G - H (iv) transpose of J (v) mean of values of G. Describe the use of an array. Write program to calculate $c = a^2 - b^2$ and d = a+ b. Implement your program using loops and arrays.	6+4
Q.4.	Write C++ program for the Damped harmonic motion (D.H.M) of a mass attached with a spring using Euler's method under the following conditions: (g=9.8 m/s2, initial position zero and velocity 15 m/s, time step 0.1 sec. and maximum time 15 sec., k = 1 N/m, m=1kg, damping coefficient = 0.5 N/ms, =0.01 s-1 and f0=1.5N.). Calculate and print values for time, position, velocity and acceleration. How you can change the same program for Simple H.M., and Forced H.M. Write C++ program which reads in a number as decimal number and convert it into binary number. Implement your program for 10 iterations.	5+5

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fifth Semester – Spring 2023

Roll No. ....

Paper: Computational Physics - I | **Course Code: PHYS-3502** | **Time: 3 Hrs.** | **Marks: 60**

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions: (6x5=30)

I. Write a C++ program to print the given series and sum of the series: $S = \sum_{n=1}^{10} \frac{(-1)^n x^{n/2}}{n(n+1)}$, ask user to input value of x.

II. Write C++ program with function to calculate $f(x) = 5x^2 - 2x - 1$ and print values for x and f(x). Implement for five iterations.

III. Write a C++ program to calculate and print factorial of a number taken from user using while loop.

IV. Write a C++ program to generate random numbers in a range of 0 to N.

V. Write a C++ program to find a trace of matrix of order 4x4.

VI. Write a C++ program to find a transpose of matrix of order 3x3.

Q.2. Answer the following questions. (3x10=30)

I. Write a C++ program to solve the differential equation $\frac{dy}{dx} = x^3 - y$ subject to the initial condition $y(0)=1$ and $h=0.05$ on the interval [0,0.2].

II. Write a C++ program to use the Trapezoidal rule with n=6 to estimate the integral $I = \int_1^2 \frac{1}{x} dx$.

III. Write a C++ program to evaluate the $\frac{dy}{dx} = x + y; y(0) = 1, [0,0.5], h = 0.1$ by Runge-Kutta method (4th order).

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B.S. 4 Years Program / Fifth Semester – Fall 2022

Roll No. ...

Paper: Computational Physics - I | Course Code: PHYS-3502 | Time: 3 Hrs. | Marks: 60

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Q.1. Answer the following short questions: (6x5=30)

I. Write a C++ program to calculate momentum and kinetic energy of the particle using functions such that P = mv and K.E.= 1/2 mv².

II. Write a C++ program to print diagonal elements of C= A x B, where A and B are 3x3 matrices.

III. Write a C++ program to print the inverse of a product of two matrices of order 3x3.

IV. Write a C++ program to calculate time period and print l(length) against T for length L=0.2-2, h=0.2 and $T = 2\pi\sqrt{l/g}$.

V. Write a C++ program to calculate Area and Circumference of circle using array size=7.

VI. Write a C++ program to print a 4x3 matrix by user entered values.

Q.2. Answer the following questions. (3x10=30)

I. Write a C++ program to calculate and print time against position values for Damped Ha[illegible]nic Oscillator by using Euler's Method for: $a = -kx - {bv}/{m}$, given that $g = 9.8\, m/s^2$, [illegible]itial position=x=0, Initial velocity=v_i=50m/s, Initial time=t=0 s, h=0.15 s, t_{maxi}=5.5s, mass=[illegible]1kg, k=, b=0.5.

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UNIVERSITY OF THE PUNJAB

BS (After Associate Degree) : Fifth Semester – Fall 2022

Roll No. ....

Paper: Computational Physics – I   Course Code: PHYS-3502   Time: 3 Hrs.   Marks: 60

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Q.1. Solve the following questions. (6x5=30)

I. Explain with example the following terms in C++:
  a. float x
  b. switch()
  c. sqrt()

II. Write syntax with example for the followings in C++:
  a. do-while loop
  b. for loop

III. Write a program to print the given series and sum of the series: , ask user to input value of x.

IV. Write C++ program with function to calculate $f(x) = 20x^2 - 8x+3$ and print values for x and f(x). Implement for seven iterations.

V. Write C++ program to calculate and print factorial of a number taken from user using while loop. Implement for six iterations.

VI. Write C++ program to calculate and print equivalent resistor of seven resistors connected in series.

Q.2. Solve the following questions. (3x10=30)

I. Write a C++ program for the decay of current in a simple RL-circuit using Euler's method with initial conditions: r=10Ω, L=5H, initial time 0, time step 0.1, maximum time 2.5sec., initial current 5A and voltage v=0 volts. Print current against time values. How you can convert the same program for the growth of current in the circuit?

II. Write a program to evaluate the integral by Trapezoidal rule, I= 33.604.

III. Write a program to find a product of two matrices of order 3x3 and calculate the trace of these matrices

UNIVERSITY OF THE

BS (After Associate Degree) : Fifth Semester – Fall 2021 Time: 3 Hrs.

Paper: Computational Physics – I Course Code: PHYS-3502

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

(6x5=30)

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